1. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data?*
   (a) Statistics explain why relationship exist between variables
   (b) Statistics simplify complex information to help us make better decisions
   (c) Statistics provide honest answers to complex questions
   (d) Statistical formulas are the most important part of understanding data

2. National survey data shows that 69% of people support mandatory childhood vaccinations. A local researcher believes that support is likely much higher in Toronto compared with the national average. Which of the following equations represents the research hypothesis?
   (a) $\mu = 0.69$
   (b) $\mu < 0.69$
   (c) $\mu \neq 0.69$
   (d) $\mu > 0.69$

3. The U.S. General Social Survey asked people how many brothers and sisters they had. A researcher used R to produce the summary statistics of this variable (`sibs`) below. Based on this, you conclude the distribution for this variable is:

| freq | median | mean | sd |
|---|---|---|---|
| 3291 | 3 | 2.9 | 1.8 |

   (a) nominal
   (b) positively skewed
   (c) negatively skewed
   (d) symmetrical

4. Based on the table below, which statement best compares the relative frequency of being `not too happy` between men and women?

   Cross-Tabulation, Column Proportions
   happy * sex
   Data Frame: gss24

| sex | male | female | Total |
|---|---|---|---|
| happy | | | |
| very happy | 293 ( 20.2%) | 386 ( 21.3%) | 679 ( 20.8%) |
| pretty happy | 830 ( 57.1%) | 1054 ( 58.2%) | 1884 ( 57.7%) |
| not too happy | 330 ( 22.7%) | 370 ( 20.4%) | 700 ( 21.5%) |
| Total | 1453 (100.0%) | 1810 (100.0%) | 3263 (100.0%) |

   (a) Women have a slightly higher relative frequency because more women than men reported being `not too happy`.
   (b) The frequency of being `not too happy` is equal for men and women.
   (c) A greater number of women reported being `not too happy` than men, so women have a higher relative frequency.
   (d) Men have a slightly higher relative frequency of being `not too happy` than women, even though fewer men reported it.

5. In order to compare meaningfully across groups, a frequency should be converted into a/an ____________.
   (a) standard deviation
   (b) ordinal variable
   (c) percentage
   (d) number

6. Which description best describes a correlation coefficient of r = -0.62?
   (a) A statistically insignificant relationship
   (b) No relationship exists between the variables
   (c) Weakly negative relationship
   (d) Strong negative relationship

7. In Naked Statistics, Charles Wheelan explains that "most sample means will lie reasonably close to the population mean; the standard error is what defines 'reasonably close.' " What does this statement imply?
   (a) The standard error measures how much sample means tend to vary around the population mean.
   (b) The standard error increases as sample size increases.
   (c) The sample mean is always equal to the population mean.
   (d) The standard error tells us how far the population mean is from the sample mean.

8. Researchers calculated a 95% confidence interval of the average family size of Canadians to be from 2.7 to 3.1, based on a sample of 500 families with a standard deviation of 1.1. What change would produce a more precise estimate?
   (a) Increasing our sample size to 600
   (b) Increasing our confidence interval to 99%
   (c) Improving our sampling strategy
   (d) Increasing our standard deviation to 1.3

9. Which RStudio pane is used to write and edit R scripts and Quarto documents?
   (a) Console
   (b) Source
   (c) Files
   (d) Environment

10. According to Wheelan, what is the most common cause of inaccurate polling results?
    (a) Biased sampling methods or poorly worded questions
    (b) Misuse of statistical software during data analysis
    (c) Random variation in respondent answers
    (d) Incorrect calculations of standard errors

11. Using the U.S. General Social Survey data, a researcher tests whether the proportions who favor a law requiring a person to obtain a police permit before they could buy a gun have changed over time. Assuming an $\alpha$ level of .05, which of these statements best summarizes the researcher's conclusion?

    ```
    Welch Two Sample t-test
    ```

    data: my_data$gunlawbymy_data$year t = 2, df = 3115, p-value = 0.07 alternative hypothesis: true difference in means between group 1984 and group 2024 is not equal to 0 95 percent confidence interval: -0.0022 0.0581 sample estimates: mean in group 1984 mean in group 2024 0.72 0.70

    (a) There is not a statistically significant difference in support for the police permit law between 1984 and 2024, indicating that public opinion has remained stable.
    (b) The difference in support for the police permit law between 1984 and 2024 was due to sampling error and cannot be interpreted.
    (c) The is a statistically significant increase in support for the police permit law from 1984 to 2024, proving that stricter gun control laws are now favored by most Americans.
    (d) There is a statistically significant difference in support for the police permit law between 1984 and 2024, suggesting that public opinion has changed over time.

12. Has public opinion changed about whether Black Americans have worse jobs, income, and housing than white Americans mainly because of discrimination? Based on a significance level of $\alpha = 0.01$, what's the best conclusion from the test?

| | year | | |
|---|---|---|---|
| | 2022 | 2024 | Total |
| opinion | | | |
| no | 509 (45%) | 522 (51%) | 1,031 (48%) |
| yes | 619 (55%) | 494 (49%) | 1,113 (52%) |
| Total | 1,128 (100%) | 1,016 (100%) | 2,144 (100%) |

Pearson's Chi-squared test, p=0.004

(a) The evidence shows attitudes about racial inequality differ between 2022 and 2024.
(b) The results show that most people in both years agreed racial inequality is due to discrimination.
(c) The p-value of .000 means the null hypothesis is true.
(d) There is no statistically significant difference in attitudes between 2022 and 2024.

13. What is the purpose of `drop_na()` in dplyr?

    (a) It removes rows that contain missing values.
    (b) It replaces missing values with zeros.
    (c) It drops columns that are entirely NA.
    (d) It converts NA values to NULL.

14. Chi-square values cannot be _____.

    (a) zero
    (b) positive
    (c) negative
    (d) outside the range of 0 to 1

15. Why is it considered good practice to use RStudio Projects when working on data analysis?

    (a) They automatically load packages when you open RStudio, so there's no need to use `library()`
    (b) They prevent syntax errors by enforcing tidyverse formatting rules
    (c) They help organize your files, set a working directory, and keep your environment consistent
    (d) They ensure your code runs faster by optimizing memory usage

16. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?

    (a) This couple had about 8 arguments per month.
    (b) This couple had fewer arguments than about 2% of couples.
    (c) This couple had fewer arguments than about 98% of couples.
    (d) This couple had fewer arguments than about 5% of couples.

17. What will the following code do?

```
gss24 <- gss24 |>
  mutate(status = case_when(
  income > 100000 ~ "High",
  income > 50000 ~ "Medium",
  TRUE ~ "Low"
))
```

    (a) It filters out rows where income is less than 50,000.
    (b) It creates a new column `status` with 3 possible values.
    (c) It replaces the numeric values in the income column with text labels.
    (d) It removes missing values from the income column.

18. According to the 68-95-99.7 rule, what can we say about a person whose score is more than 2 standard deviations above the mean?
    (a) Their score is likely below average.
    (b) Their score is unusual and higher than most.
    (c) Their score is within the middle 68% of the distribution.
    (d) Their score is very common.

19. A researcher used the 2024 General Social Survey to find the proportion of the U.S. population that reports believing in some form of life after death. Based on the R output of their findings, with 95% confidence intervals, which of the following is a correct interpretation of their results?

| sex | n | mean | sd | lowCI | hiCI |
|---|---|---|---|---|---|
| male | 1139 | 4.2 | 3.7 | 4.0 | 4.5 |
| female | 1174 | 4.0 | 3.5 | 3.8 | 4.2 |

    (a) We cannot determine whether the null hypothesis of zero difference between the parameters will be rejected at the 0.05 level.
    (b) The confidence intervals overlap, so we can't know for sure if there is a statistical difference between the average time men and women spend on leisure activities per day.
    (c) We are 95% certain that women have less time per day than men for leisure activities, likely because of their greater time spent on housework and childcare.
    (d) The confidence intervals do not overlap so we can conclude that the average time men and women spend on leisure activities per day are statistically significantly different.

20. Which function is used to load a package in R?
    (a) `install.packages()`
    (b) `load()`
    (c) `source()`
    (d) `library()`

21. If a researcher concluded that there was no difference in police use of force against Black and White men when there actually is a difference, then:
    (a) The null hypothesis was incorrectly rejected (false positive).
    (b) The null hypothesis was correctly rejected.
    (c) The null hypothesis was incorrectly retained (false negative).
    (d) The null hypothesis was correctly retained.

22. A researcher compares support for universal healthcare (Support vs. Oppose) between two racial groups: Black and White respondents. Which statistical test should be used?
    (a) Linear regression
    (b) Chi-square test
    (c) Pearson's correlation (r)
    (d) t-test

23. What will happen if you try to use a function from a package that hasn't been loaded?
    (a) The function will run normally
    (b) R will prompt you to install the package
    (c) R will automatically load the package
    (d) You will get an error saying the function is not found

24. According to Cohen's argument, what role should descriptive statistics play in social science research?
    (a) Descriptive statistics are outdated and should be replaced by more advanced inferential methods.
    (b) Descriptive work should be minimized in favor of studies that make strong theoretical contributions.
    (c) Descriptive statistics are a foundational part of scientific inquiry and should be valued as legitimate research.
    (d) Descriptive statistics are useful only as a predominately as a step before hypothesis testing and theory development.

25. What does Oscar Gandy suggest about the use of racial statistics?
    (a) Their use in legal proceedings increasingly lead to unfair policy outcomes
    (b) They are neutral tools most useful for academic research
    (c) They can both expose and perpetuate discrimination
    (d) Their use in data journalism helps mitigate online polarization

26. The level of measurement for the `hapmar` variable is _________

    [1] "Happiness of marriage" Frequencies
    gss24$hapmar
    Type: Factor

| | Freq | % | % Cum. |
|---|---|---|---|
| very happy | 802 | 59.36 | 59.36 |
| pretty happy | 504 | 37.31 | 96.67 |
| not too happy | 45 | 3.33 | 100.00 |
| Total | 1351 | 100.00 | 100.00 |

    (a) nominal
    (b) interval
    (c) ordinal
    (d) dichotomous

27. The variable `mntlhlth` measures how many days in the past 30 a respondent reported poor mental health. A respondent reports 5 poor mental health days. Based on the summary statistics for this variable, which of the following best describes their z-score?

| freq | median | mean | sd |
|---|---|---|---|
| 2279 | 0 | 4.5 | 7.9 |

    (a) The z-score is undefined because the median is 0.
    (b) The z-score is large and positive because 5 is much higher than the mean.
    (c) The z-score is negative because 5 is below the sd.
    (d) The z-score is close to zero because 5 is close to the mean.

28. In a normally distributed set of test scores, a student scoring below the average will have a _______________.
    (a) negative z-score
    (b) positive standard error
    (c) negative standard error
    (d) positive z-score

29. Using the U.S. General Social Survey data, a researcher suspects Americans are watching more tv today than in the 1970s. Assuming an $\alpha$ level of .05, should the researcher reject the null hypothesis? Why or why not?

```
Welch Two Sample t-test
```

data: my_data$tvhoursbymy_{d}ata$year t = -3, df = 3617, p-value = 0.005 alternative hypothesis: true difference in means between group 1975 and group 2024 is not equal to 0 95 percent confidence interval: -0.432 -0.079 sample estimates: mean in group 1975 mean in group 2024 3.0 3.3

(a) No, because the p-value (0.005) is greater than the $\alpha$ level of .05.
(b) Yes, because the p-value (0.005) is less than the $\alpha$ level of .05.
(c) Yes, because the p-value (0.005) is greater than the $\alpha$ level of .05.
(d) No, because the p-value (0.005) equals the $\alpha$ level of .05.

30. What does the following R code produce?

```
my_data |>
  tbl_cross(
    row = incgap,
    col = sex,
    percent = "column",
    missing = "no") |>
  add_p(source_note = TRUE)
```

(a) A cross-tabulation of `incgap` by `sex`, showing column percentages and a chi-square test of independence
(b) A regression output predicting `incgap` from `sex`, including R-squared and p-values
(c) A frequency table of `incgap` responses with missing values imputed and grouped by `sex`
(d) A summary table comparing the mean values of `incgap` between men and women using a t-test

31. Which of the following is a valid use of `group_by()` and `summarize()`?

(a) summarize(group_by(data, age), mean(sex))
(b) group_by(data, summarize(sex))
(c) data |> summarize(mean = group_by(sex))
(d) data |> group_by(sex) |> summarize(freq = n())

32. What is the result of the following code?

```
gss24 |>
  filter(age > 20) |>
  select(age, childs)
```

(a) Only rows with `age` > 20 and columns `age` and `childs` are returned.
(b) A new column is added to `gss24`
(c) rows with `age` > 20 are removed
(d) The original gss24 dataset is overwritten.

33. Using the table, which of the following is the correct formula to find the expected frequency for U.S. liberals who oppose the death penalty in 2024?

| | ideology | conservative | liberal | Total |
|---|---|---|---|---|
| cappun | | | | |
| favor | | 512 (66.8%) | 255 (33.2%) | 767 (100.0%) |
| oppose | | 168 (32.7%) | 345 (67.3%) | 513 (100.0%) |
| Total | | 680 (53.1%) | 600 (46.9%) | 1280 (100.0%) |

(a) (345 * 513)/1280
(b) (600 * 513)/1280
(c) (345 * 600)/1280
(d) (600 * 767)/1280

34. A p-value is
    (a) An estimate of where our population statistic lies on the true sampling distribution of the means
    (b) A measure of how unusual or rare our obtained statistic is compared with what is stated in the null hypothesis
    (c) A measure that tells us the variability in a normal distribution
    (d) An estimate of the standard error of a sampling distribution

35. What does the following code return?

```
gss24 |>
  group_by(sex) |>
  summarize(mean_age = mean(age, na.rm = TRUE))
```

    (a) A filtered data frame with only rows where age is not missing.
    (b) A new column named `mean_age` added to the original data frame.
    (c) a list of all ages grouped by sex.
    (d) a data frame showing the average age for each sex.

36. What key concept does Wheelan illustrate in the chapter titled Inference: Why My Statistics Professor Thought I Might Have Cheated?
    (a) That unusually accurate results can raise suspicion, but statistical inference helps assess whether they are likely due to chance
    (b) That cheating can be detected by comparing mean scores across different groups
    (c) That statistical inference can be used to prove someone's guilt beyond reasonable doubt
    (d) That hypothesis testing is not useful in criminal investigations

37. A sociologist wants to examine whether there is a relationship between level of education (measured in years) and annual income (in dollars) among adults in the U.S. Which statistical test is most appropriate?
    (a) Logistic regression
    (b) t-test
    (c) Chi-square test
    (d) Pearson's correlation (r)

38. What does the pipe operator (|>) do in dplyr?
    (a) It passes the result of one function into the next.
    (b) It loads the dplyr package into memory.
    (c) It filters rows based on a condition.
    (d) It creates a new column in a data frame.

39. Which of the following best describes the differences between the Console, R Scripts, and Quarto documents in RStudio?
    (a) The Console and R Scripts are identical, while Quarto is used only for Markdown formatting.
    (b) R Scripts and Quarto documents both run code line-by-line, but only the Console saves your work.
    (c) The Console runs code interactively, R Scripts store reusable code, and Quarto combines code, text, and output for dynamic reports.
    (d) The Console is used for writing long-term code, R Scripts are for temporary testing, and Quarto is only for plotting.

40. What does the expression `5 != 3` evaluate to in R?
    (a) NULL
    (b) NA
    (c) FALSE
    (d) TRUE

41. Which command installs a new package in R?
    (a) `install.packages()`
    (b) `load()`
    (c) `require()`
    (d) `library()`
42. If a statistics classes' final grades were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?
    (a) 50%
    (b) 62%
    (c) 95%
    (d) 68%
43. What does the `::` operator do in R?
    (a) Assigns a value to a variable
    (b) Specifies a function from a particular package (without loading it)
    (c) Comments out a line of code
    (d) Imports all functions from a package
44. Which of the following statements best reflects the ASA's guidance on interpreting p-values?
    (a) P-values measure the scientific or economic significance of a result.
    (b) A small p-value always indicates a large and important effect.
    (c) A large p-value proves that no effect exists.
    (d) P-values are influenced by sample size and measurement precision, and do not reflect effect size or importance.
45. Which of the following code snippets correctly uses mutate() to convert height from centimeters to meters?
    (a) mutate(height / 100 = height)
    (b) mutate(height = select(height / 100))
    (c) mutate(height <- height / 100)
    (d) mutate(height = height / 100)
46. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?
    (a) cumulative frequencies
    (b) frequencies
    (c) percentages
    (d) cumulative percentages
47. Which description best describes the correlation between personal income (`realrinc`) and number of days of poor mental health (`mntlhlth`)?

    ```
    Pearson's product-moment correlation
    ```

    data: realrinc and mntlhlth t = -5, df = 1933, p-value = 5e-07 alternative hypothesis: true correlation is not equal to 0 95 percent confidence interval: -0.16 -0.07 sample estimates: cor -0.11

    (a) Results indicated an inverse relationship between personal income and days of poor mental health (r = -0.11, p < .05), suggesting that the number of days of poor mental health decreases as personal income increases.
    (b) There is no evidence of an association between personal income and days of poor mental health (t = -5).
    (c) Decreases in personal income was correlated with decreases in the number of days of poor mental health (t = -5, p < .05).
    (d) There is a weakly positive association between personal income and days of poor mental health (r = -0.11, p < .05).

48. Which of the following is a measure of the spread of a distribution?

(a) mean
(b) standard deviation
(c) standardized variable
(d) median

49. Which of these datasets has a standard deviation of 0?

(a) 0, 2, 4, 6, 8
(b) 1, 4, 7, 10, 15
(c) 4, 4, 4, 4, 4
(d) 2, 2, 5, 7, 7

50. A researcher used the 2024 General Social Survey to find the proportion of the U.S. population that reports believing in some form of life after death. Based on the R output with 90% confidence intervals, which of the following is a correct interpretation of their results?

| n | mean | sd | lowCI | hiCI |
|---|---|---|---|---|
| 1283 | 0.83 | 0.38 | 0.81 | 0.85 |

(a) Because the mean is within the confidence interval, there is a .90 probability that the proportion of U.S. adults who believe in some form of life after death is .83.
(b) About 90% of people responded to the national survey.
(c) The researcher is 90% confident that the proportion of U.S. adults who believe in some form of life after death is between .81 and .85.
(d) If another survey had been conducted in 2024, there's a 90% chance the point estimate would be between .81 and .85.

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| 1. | | **X** | | |
| 2. | | | | **X** |
| 3. | | | **X** | |
| 4. | | | | **X** |
| 5. | | | **X** | |
| 6. | | | | **X** |
| 7. | **X** | | | |
| 8. | **X** | | | |
| 9. | | **X** | | |
| 10. | **X** | | | |
| 11. | **X** | | | |
| 12. | **X** | | | |
| 13. | **X** | | | |
| 14. | | | **X** | |
| 15. | | | **X** | |
| 16. | | | **X** | |
| 17. | | **X** | | |
| 18. | | **X** | | |
| 19. | | **X** | | |
| 20. | | | | **X** |
| 21. | | | **X** | |
| 22. | | **X** | | |

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| 23. | | | | X |
| 24. | | | X | |
| 25. | | | X | |
| 26. | | | X | |
| 27. | | | | X |
| 28. | X | | | |
| 29. | | X | | |
| 30. | X | | | |
| 31. | | | | X |
| 32. | X | | | |
| 33. | | X | | |
| 34. | | X | | |
| 35. | | | | X |
| 36. | X | | | |
| 37. | | | | X |
| 38. | X | | | |
| 39. | | | X | |
| 40. | | | | X |
| 41. | X | | | |
| 42. | | | | X |
| 43. | | X | | |
| 44. | | | | X |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 45. | (a) | | (b) | | (c) | | (d) | **X** |
| 46. | (a) | | (b) | | (c) | | (d) | **X** |
| 47. | (a) | **X** | (b) | | (c) | | (d) | |
| 48. | (a) | | (b) | **X** | (c) | | (d) | |
| 49. | (a) | | (b) | | (c) | **X** | (d) | |
| 50. | (a) | | (b) | | (c) | **X** | (d) | |